Problem Set 4

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Due: November 27, 2025

Instructions

- Please show your work! You may lose points by simply writing in the answer. If the problem requires you to execute commands in `R`, please include the code you used to get your answers. Please also include the `.R` file that contains your code. If you are not sure if work needs to be shown for a particular problem, please ask.
- Your homework should be submitted electronically on GitHub.
- This problem set is due before 23:59 on Thursday November 27, 2025. No late assignments will be accepted.

Question 1: Economics

In this question, use the `prestige` dataset in the `car` library. First, run the following commands:

```
install.packages(car)
library(car)
data(Prestige)
help(Prestige)
```

We would like to study whether individuals with higher levels of income have more prestigious jobs. Moreover, we would like to study whether professionals have more prestigious jobs than blue and white collar workers.

(a) Create a new variable `professional` by recoding the variable `type` so that professionals are coded as 1, and blue and white collar workers are coded as 0 (Hint: `ifelse`).

```
Prestige["professional"] <- NA
Prestige$professional <- ifelse(Prestige$type == "prof", 1, 0)
```

(b) Run a linear model with `prestige` as an outcome and `income`, `professional`, and the interaction of the two as predictors (Note: this is a continuous × dummy interaction.)

```
model1 <- lm(prestige ~ income + professional + income:professional,
```

Table 1: Table of Coefficients

	Dependent variable:
	Prestige Model 1
Income	0.003^{***}
	(0.0005)
Professional	37.781^{***}
	(4.248)
Income:Professional	-0.002^{***}
	(0.001)
Constant	21.142^{***}
	(2.804)
Observations	98
R^2	0.787
Adjusted R^2	0.780
Residual Std. Error	8.012 (df = 94)
F Statistic	115.878^{***} (df = 3; 94)
Note:	*p<0.1; **p<0.05; ***p<0.01

(c) Write the prediction equation based on the result.

The prediction equation based on this result is y = 21.142 + 0.003x_1 + 37.781D_1 - 0.002$_{int,D1}D_1x_1$. More specifically, for this particular model, prestige = 21.142 + 0.003*income + 37.781*professional - 0.002*income:professional.

(d) Interpret the coefficient for `income`.

The coefficient for income indicates that, on average, when professional is 0 and when the interaction between income and professional is 0, for every one unit increase in income, there is a 0.003 increase in prestige. This can also be interpreted as, on average, for blue and white collar workers, for every one unit increase in income, there is a 0.003 increase in Pineo-Porter prestige score for occupation.

(e) Interpret the coefficient for `professional`.

The coefficient for professional indicates that, on average, when income is held constant, for a one unit increase in professional, there is a 37.781 increase in prestige. This can also be interpreted as, on average, when income is held constant, for a professional, there is a 37.781 increase in Pineo-Porter prestige score for occupation.

(f) What is the effect of a $1,000 increase in income on prestige score for professional occupations? In other words, we are interested in the marginal effect of income when the variable `professional` takes the value of 1. Calculate the change in $\hat{y}$ associated with a $1,000 increase in income based on your answer for (c).

```
21.142 + 0.003*(1000) + 37.781*(1) - 0.002*(1000*1)
```

The effect of a $1,000 increase in income on prestige score for professional occupations is 59.923.

(g) What is the effect of changing one's occupations from non-professional to professional when her income is $6,000? We are interested in the marginal effect of professional jobs when the variable `income` takes the value of $6,000$. Calculate the change in $\hat{y}$ based on your answer for (c).

```
non_prof <- 21.142 + 0.003*(6000)
prof <- 21.142 + 0.003*(6000) + 37.781*(1) - 0.002*(6000*1)
prof - non_prof
```

The effect of changing one's occupation from a non-professional to professional with an income of $6,000 is a 25.781 increase in prestige score.

Question 2: Political Science

Researchers are interested in learning the effect of all of those yard signs on voting preferences.[1] Working with a campaign in Fairfax County, Virginia, 131 precincts were randomly divided into a treatment and control group. In 30 precincts, signs were posted around the precinct that read, "For Sale: Terry McAuliffe. Don't Sellout Virgina on November 5."

Below is the result of a regression with two variables and a constant. The dependent variable is the proportion of the vote that went to McAuliff's opponent Ken Cuccinelli. The first variable indicates whether a precinct was randomly assigned to have the sign against McAuliffe posted. The second variable indicates a precinct that was adjacent to a precinct in the treatment group (since people in those precincts might be exposed to the signs).

Impact of lawn signs on vote share

Precinct assigned lawn signs (n=30)	0.042
	(0.016)
Precinct adjacent to lawn signs (n=76)	0.042
	(0.013)
Constant	0.302
	(0.011)

Notes: R^2=0.094, N=131

(a) Use the results from a linear regression to determine whether having these yard signs in a precinct affects vote share (e.g., conduct a hypothesis test with $\alpha = .05$).

$H_0 : \beta_1 = 0$, $H_a : \beta_1 \neq 0$.

```r
beta1 <- 0.042
SE1 <- 0.016
n1 <- 30
TS1 <- (beta1 - 0)/SE1
p_value1 <- 2*pt(abs(TS1), n1-1, lower.tail = F)
```

We can reject the null hypothesis that having yard signs in a precinct does not effect vote share, due to the fact that the p-value is 0.013, which is smaller than the alpha value of 0.05, making the effect of 0.042 on vote share significant.

[1] Donald P. Green, Jonathan S. Krasno, Alexander Coppock, Benjamin D. Farrer, Brandon Lenoir, Joshua N. Zingher. 2016. "The effects of lawn signs on vote outcomes: Results from four randomized field experiments." Electoral Studies 41: 143-150.

(b) Use the results to determine whether being next to precincts with these yard signs affects vote share (e.g., conduct a hypothesis test with $\alpha = .05$).

$H_0 : \beta_2 = 0$, $H_a : \beta_2 \neq 0$.

```
beta2 <- 0.042
SE2 <- 0.013
n2 <- 76
TS2 <- (beta2 - 0)/SE2
p_value2 <- 2*pt(abs(TS2), n2-1, lower.tail = F)
```

We can reject the null hypothesis that being next to precincts with yard signs does not effect vote share, due to the fact that the p-value is 0.002, which is smaller than the alpha value of 0.05, making the effect of 0.042 on vote share significant.

(c) Interpret the coefficient for the constant term substantively.

When there are no lawn signs utilized, the vote share will be equal to 0.302.

(d) Evaluate the model fit for this regression. What does this tell us about the importance of yard signs versus other factors that are not modeled?